OLS: Estimation, Properties, and Inference

The sample analog of the BLP — and what we can learn from it

Gov 2001 · Scott Cunningham · Spring 2026

The BLP is a population object — OLS is how we estimate it from data

Population BLP (estimand)

$$\beta = \arg\min_b \mathbb{E}\left[(Y - X'b)^2\right]$$

Defined by population moments.
We can never observe it directly.

$\Longrightarrow$

OLS (estimator)

$$\hat{\beta} = \arg\min_b \frac{1}{n}\sum_{i=1}^{n}(Y_i - X_i'b)^2$$

Same problem, sample moments replace population moments.

OLS is a **plug-in estimator**: replace $\mathbb{E}[\,\cdot\,]$ with $\frac{1}{n}\sum_i$ and solve the same FOC

OLS minimizes the sum of squared residuals — the FOC give the normal equations

Minimize over b_0, b_1 (bivariate for clarity):

$$\min_{b_0,\, b_1} \sum_{i=1}^{n} \left(Y_i - b_0 - b_1 X_i\right)^2$$

First-order conditions (set derivatives to zero):

$$\sum_{i=1}^{n} \hat{e}_i = 0 \qquad \sum_{i=1}^{n} X_i\, \hat{e}_i = 0 \qquad \text{where } \hat{e}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$$

$$\boxed{\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\widehat{\mathrm{Cov}}(X, Y)}{\widehat{\mathrm{Var}}(X)}}$$

The sample covariance-to-variance ratio — the exact sample analog of the population BLP slope.

In matrix form: $\hat{\beta} = (X'X)^{-1}X'Y$

Stack the data: $Y = X\beta + \varepsilon, \quad X$ is $n \times k, \quad Y$ is $n \times 1$

OLS solution:

$$\boxed{\hat{\beta} = (X'X)^{-1}X'Y}$$

Two requirements:

- $(X'X)$ must be invertible
- No perfect multicollinearity among columns of X

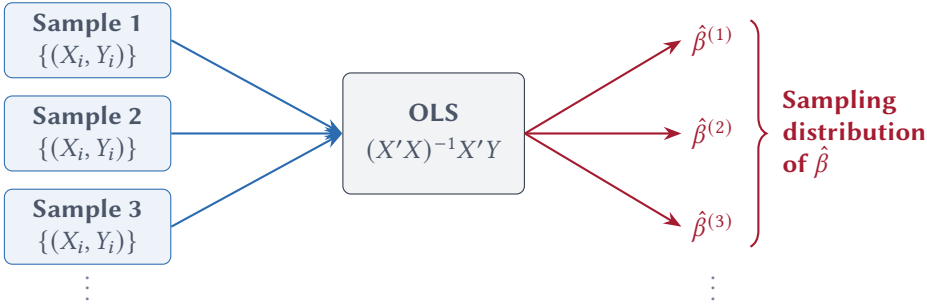
Residuals by construction:

$$\hat{e} = Y - X\hat{\beta}$$

$$X'\hat{e} = 0 \quad \text{(FOC, exactly)}$$

$\hat{\beta}$ is a function of the data (X, Y) — a random variable with its own sampling distribution.

OLS is a random variable — it has a sampling distribution



Each dataset gives a different $\hat{\beta}$. The distribution of $\hat{\beta}$ across datasets is what we analyze.

OLS is unbiased when $\mathbb{E}[\varepsilon \mid X] = 0$

Decompose $\hat{\beta}$:

$$\hat{\beta} = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + \varepsilon) = \beta + (X'X)^{-1}X'\varepsilon$$

Take expectations conditional on X:

$$\mathbb{E}[\hat{\beta} \mid X] = \beta + (X'X)^{-1}X' \underbrace{\mathbb{E}[\varepsilon \mid X]}_{=0 \text{ by assumption}} = \beta$$

$$\boxed{\mathbb{E}[\hat{\beta} \mid X] = \beta \quad \text{— OLS is } \textbf{unbiased} \text{ under strict exogeneity}}$$

Finite-sample result: holds for any n, not just large n. The assumption $\mathbb{E}[\varepsilon_i \mid X_1, \ldots, X_n] = 0$ is stronger than what consistency needs.

Unbiasedness vs. consistency — two different properties, two different standards

Unbiased: $\mathbb{E}[\hat{\beta}] = \beta$

Centered on the truth *on average* across repeated samples

Finite-sample property. Holds for any n.

Consistent: $\hat{\beta} \xrightarrow{p} \beta$

Converges to the truth as $n \to \infty$

Asymptotic property. Requires LLN on moments.

- An estimator can be **biased but consistent** — e.g., MLE of σ^2 with n in denominator
- An estimator can be **unbiased but inconsistent** — e.g., Y_1 alone always targets μ but never converges
- OLS is **both** under standard assumptions

Unbiasedness is a stronger assumption. Consistency is weaker but what large samples guarantee.

OLS is consistent under a weaker condition — only $\mathbb{E}[X_i\varepsilon_i] = 0$ required

Why OLS is consistent:

$$\hat{\beta} - \beta = \left(\frac{1}{n}\sum_i X_i X_i'\right)^{-1} \frac{1}{n}\sum_i X_i \varepsilon_i$$

By LLN:

$$\frac{1}{n}\sum_i X_i X_i' \xrightarrow{p} \mathbb{E}[X_i X_i'] \equiv Q$$

$$\frac{1}{n}\sum_i X_i \varepsilon_i \xrightarrow{p} \mathbb{E}[X_i \varepsilon_i] = 0$$

By Slutsky:

$$\hat{\beta} - \beta \xrightarrow{p} Q^{-1} \cdot 0 = 0$$

$$\therefore \quad \hat{\beta} \xrightarrow{p} \beta$$

Consistency requires only $\mathbb{E}[X_i\varepsilon_i] = 0$ — a moment condition, not $\mathbb{E}[\varepsilon \mid X] = 0$

Unbiasedness needs the full $\mathbb{E}[\varepsilon \mid X] = 0$ (strict exogeneity). Consistency needs only the weaker version.

CLT gives normality to sample means — OLS is not a sample mean

CLT applies directly to simple averages:

$$\sqrt{n}\,(\bar{X} - \mu) \xrightarrow{d} N(0,\ \mathrm{Var}(X))$$

OLS is not a simple average:

$$\hat{\beta} = (X'X)^{-1}X'Y$$

- A ratio of matrix sums — nonlinear in the data
- CLT does not apply *directly* to $\hat{\beta}$
- Saying OLS is asymptotically normal is asserting something nontrivial

"Isn't everything normal under CLT?" — Only simple averages get normality for free. OLS requires showing it reduces to a scaled average. That's the content of the theorem.

OLS is approximately a scaled sample mean — that's why CLT applies

Decompose the estimation error:

$$\hat{\beta} - \beta = \underbrace{\left(\frac{X'X}{n}\right)^{-1}}_{\xrightarrow{p} Q^{-1} \text{ (LLN)}} \cdot \underbrace{\frac{X'\varepsilon}{n}}_{\text{sample mean of } X_i\varepsilon_i}$$

CLT on the numerator:

$$\sqrt{n} \cdot \frac{1}{n} \sum_i X_i \varepsilon_i \xrightarrow{d} N(0,\ \Sigma) \qquad \Sigma = \mathbb{E}[X_i X_i' \varepsilon_i^2]$$

Slutsky handles the denominator: multiply by $Q^{-1} \xrightarrow{p} Q^{-1}$

$$\boxed{\sqrt{n}\,(\hat{\beta} - \beta) \xrightarrow{d} N\big(0,\ Q^{-1}\Sigma Q^{-1}\big)}$$

CLT does the heavy lifting — but on $X_i\varepsilon_i$, not on Y_i directly. LLN stabilizes the denominator. Slutsky combines them.

Asymptotic normality is the foundation for every piece of inference we do

What $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V)$ gives us:

For large n:

$$\hat{\beta} \approx N\left(\beta, \frac{V}{n}\right)$$

We know the shape of the sampling distribution — even though β is unknown.

What we still need:

An estimate $\hat{V}$ of V

$\hat{V}$ determines our standard errors — and getting it wrong breaks all inference.

The question "what is V?" is exactly the homoskedasticity vs. heteroskedasticity question — coming up next.

Gauss-Markov: OLS is BLUE under four assumptions

The four classical assumptions:

1. **Linearity:** $Y_i = X_i'\beta + \varepsilon_i$
2. **Full rank:** $(X'X)$ invertible — no perfect multicollinearity
3. **Strict exogeneity:** $\mathbb{E}[\varepsilon_i \mid X_1, \ldots, X_n] = 0$
4. **Homoskedasticity:** $\text{Var}(\varepsilon_i \mid X) = \sigma^2$ for all i

Gauss-Markov Theorem: Under (1)–(4), OLS is **BLUE** — Best Linear Unbiased Estimator

Among all estimators of the form $\tilde{\beta} = AY$ that satisfy $\mathbb{E}[\tilde{\beta}] = \beta$, OLS achieves the smallest variance for every coefficient.

BLUE: what each word means

B	L	U	E
Best	**Linear**	**Unbiased**	**Estimator**
Minimum variance	$\tilde{\beta} = AY$ a linear function of the data	$\mathbb{E}[\tilde{\beta}] = \beta$ centered on the truth	$\hat{\beta}_k$, one component at a time

- "Best among linear unbiased" — not best among *all* estimators
- Finite-sample result: no asymptotic argument needed
- **Requires homoskedasticity (assumption 4)** — this is the critical one
- Under heteroskedasticity: OLS is still unbiased, but **no longer best**

WLS (weighted least squares) is BLUE under heteroskedasticity. But we usually just fix the SEs.

What you get — and what you need — under each assumption

Assumptions held	OLS property	What breaks without it
$\mathbb{E}[\varepsilon \mid X] = 0$	Unbiased	Point estimates are off
+ Homoskedasticity	BLUE (Gauss-Markov)	OLS is inefficient; WLS beats it
+ Large n (LLN, CLT)	Consistent + asymp. normal	No sampling distribution
Heterosk. only	Unbiased, consistent, normal	**Standard errors are wrong**

Heteroskedasticity does not bias $\hat{\beta}$ — it corrupts our estimate of $\text{Var}(\hat{\beta})$

This is why heteroskedasticity is fundamentally an inference problem, not an estimation problem.

The variance of $\hat{\beta}$ is the pivot of all inference

From asymptotic normality: $\hat{\beta}_k \approx N\Big(\beta_k,\ \mathrm{Var}(\hat{\beta}_k)\Big)$

Every inferential object depends on $\mathrm{Var}(\hat{\beta}_k)$**:**

Standard error	t-statistic	Confidence interval
$\mathrm{SE}(\hat{\beta}_k) = \sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_k)}$	$t = \dfrac{\hat{\beta}_k - \beta_k^0}{\mathrm{SE}(\hat{\beta}_k)}$	$\hat{\beta}_k \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_k)$

Wrong $\widehat{\mathrm{Var}}(\hat{\beta}_k)$ $\Rightarrow$ wrong SE $\Rightarrow$ wrong t $\Rightarrow$ wrong p-values and CIs

Getting the variance formula right is not a detail. It is the whole inference problem.

Homoskedasticity: one number summarizes the error variance everywhere

Assumption: $\text{Var}(\varepsilon_i \mid X_i) = \sigma^2$ (constant, does not depend on X_i)

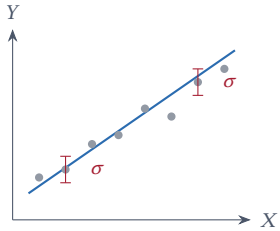
Population variance of $\hat{\beta}$:

$$\text{Var}(\hat{\beta}) = \sigma^2 (X'X)^{-1}$$

Estimated:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$$

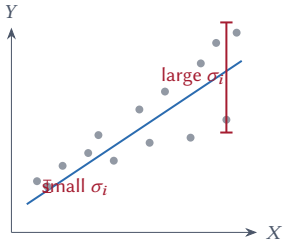
$$\hat{\sigma}^2 = \frac{1}{n-k} \sum_{i=1}^{n} \hat{e}_i^2$$



One number $\hat{\sigma}^2$ scales the whole variance matrix. Clean — but only valid under homoskedasticity.

Heteroskedasticity: the error variance changes with X — the homoskedastic formula breaks

Fan-shaped residuals:



$\text{Var}(\varepsilon_i \mid X_i) = \sigma_i^2$ **varies with** X_i

The homoskedastic formula uses $\hat{\sigma}^2$ **— a single average:**

$$\widehat{\text{Var}}_{hom}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1}$$

- Over-smooths the true variance structure
- Wrong in every part of the X range
- SEs too small where variance is large
- SEs too large where variance is small

Result: t-statistics, p-values, and CIs are all wrong.

The sandwich estimator lets each residual speak for itself

Eicker-Huber-White (robust) variance estimator:

$$\widehat{\text{Var}}_{robust}(\hat{\beta}) = \underbrace{(X'X)^{-1}}_{\text{bread}} \underbrace{\left(\sum_{i=1}^{n} \hat{e}_i^2 \, X_i X_i'\right)}_{\text{meat}} \underbrace{(X'X)^{-1}}_{\text{bread}}$$

Why it works:

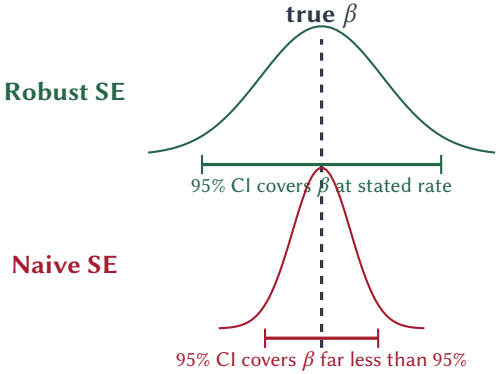
- $\hat{e}_i^2$ estimates σ_i^2 at each observation
- Weights $X_i X_i'$ by the local variance
- No assumption on the shape of σ_i^2

In practice:

- `lm_robust()` in R (`estimatr`)
- `vce(robust)` in Stata
- Use by default — the cost of not using it can be severe

Angrist & Pischke: "always report heteroskedasticity-robust standard errors." Homoskedasticity is an assumption; robust SEs are not.

What heteroskedasticity does to inference — a visual



The estimator $\hat{\beta}$ is in the same place — unbiasedness is intact. But the spread we *report* for it is wrong. That is the entire problem.

t-statistic and confidence interval for a single coefficient $\hat{\beta}_k$

From asymptotic normality:

$$\frac{\hat{\beta}_k - \beta_k}{\mathrm{SE}(\hat{\beta}_k)} \xrightarrow{d} N(0,1)$$

Hypothesis test ($H_0 : \beta_k = 0$):

$$t = \frac{\hat{\beta}_k}{\mathrm{SE}(\hat{\beta}_k)}$$

Reject H_0 at 5% if $|t| > 1.96$

p-value $= 2\mathbb{P}(Z > |t|)$, $Z \sim N(0,1)$

95% confidence interval:

$$\hat{\beta}_k \pm 1.96 \cdot \mathrm{SE}(\hat{\beta}_k)$$

Covers the true β_k in 95% of repeated samples

Use t_{n-k} critical values for small n

Same logic as the inference lectures — now $\mathrm{SE}(\hat{\beta}_k) = \sqrt{\widehat{\mathrm{Var}}(\hat{\beta}_k)/n}$

Robust or not robust: the formula is the same. What changes is how we estimate $\mathrm{Var}(\hat{\beta}_k)$.

F-test: testing joint hypotheses on multiple coefficients

Why not just run multiple t-tests?

- Testing $\beta_1 = 0$ and $\beta_2 = 0$ separately inflates the Type I error rate
- A joint test holds the false rejection rate at α for the *joint* null

Restricted vs. unrestricted models:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n-k)} \sim F_{q,\,n-k} \text{ under } H_0$$

- SSR_R: sum of squared residuals imposing H_0
- SSR_U: sum of squared residuals of the full model
- q: number of restrictions
- Intuition: how much does imposing H_0 hurt the fit?

Special case: $q = 1 \Rightarrow F = t^2$. The F-test is the generalization of the t-test to multiple restrictions.

F-test example: are all slope coefficients jointly zero?

Null: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_{k-1} = 0$

Restricted model (intercept only):

$$Y_i = \beta_0 + \varepsilon_i$$

$$SSR_R = \sum (Y_i - \bar{Y})^2 = TSS$$

Unrestricted model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \varepsilon_i$$

$$SSR_U = \sum \hat{e}_i^2$$

$$F = \frac{(TSS - SSR_U)/(k-1)}{SSR_U/(n-k)} = \frac{R^2/(k-1)}{(1-R^2)/(n-k)}$$

When F is large, the regressors together explain enough variance to reject H_0

This is the "overall F-statistic" reported by every regression package. It answers: does the model explain anything at all?

R^2 is the fraction of Y-variance explained by the regression

Variance decomposition:

$$\underbrace{\sum_i (Y_i - \bar{Y})^2}_{TSS} = \underbrace{\sum_i (\hat{Y}_i - \bar{Y})^2}_{ESS} + \underbrace{\sum_i \hat{e}_i^2}_{SSR}$$

$$\boxed{R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS} \in [0, 1]}$$

- $R^2 = 0$: regression explains nothing (flat line through $\bar{Y}$)
- $R^2 = 1$: perfect fit, all points on the line
- Bivariate OLS: $R^2 = \widehat{\text{Corr}}(X, Y)^2$

Adding *any* regressor — even noise — can only increase R^2 or leave it flat. Never decreases.

Adjusted R^2 penalizes complexity — $\bar{R}^2$ can fall when you add irrelevant variables

Adjusted R^2:

$$\bar{R}^2 = 1 - \frac{SSR/(n-k)}{TSS/(n-1)} = 1 - (1 - R^2)\frac{n-1}{n-k}$$

What it does:

- Divides by degrees of freedom, not raw sums
- Penalizes adding regressors that don't reduce SSR enough
- $\bar{R}^2$ *can* decrease — signals the new variable wasn't worth it

When $\bar{R}^2$ falls:

$$\Delta\bar{R}^2 < 0 \iff F_{\text{new var}} < 1$$

The new variable's F-statistic is less than 1 — it actively hurts the adjusted fit.

$\bar{R}^2$ is a better model-comparison tool than R^2 — but neither is a causal object.

What R^2 tells you — and what it doesn't

R^2 does tell you:

- How well the line fits this sample
- How much variance is explained by the included regressors
- Whether two models fit differently (via $\bar{R}^2$)

R^2 does NOT tell you:

- Whether $\hat{\beta}$ is unbiased or consistent
- Whether the causal interpretation is valid
- Whether the standard errors are correct
- Whether you should add more variables

Low R^2 with correct specification beats high R^2 with the wrong model

Causal inference cares about the specification, not the fit. $R^2 = 0.04$ with a valid instrument beats $R^2 = 0.90$ with endogenous regressors.

Today: OLS as an estimator with a sampling distribution — next: robust SE applications

The arc of today:

- **OLS** is the sample plug-in for the BLP: $\hat{\beta} = (X'X)^{-1}X'Y$
- **Unbiased** under strict exogeneity — finite-sample result
- **Consistent** under the weaker moment condition $\mathbb{E}[X_i\varepsilon_i] = 0$
- **Asymptotically normal** because $\frac{X'\varepsilon}{n}$ is a sample mean — CLT + Slutsky
- **BLUE** (Gauss-Markov) under homoskedasticity — loses "B" under heteroskedasticity
- **Heteroskedasticity** is an inference problem: use the sandwich estimator
- **Inference**: t-stats and CIs for single $\hat{\beta}_k$; F-tests for joint hypotheses
- R^2 measures fit, not correctness — $\bar{R}^2$ penalizes complexity